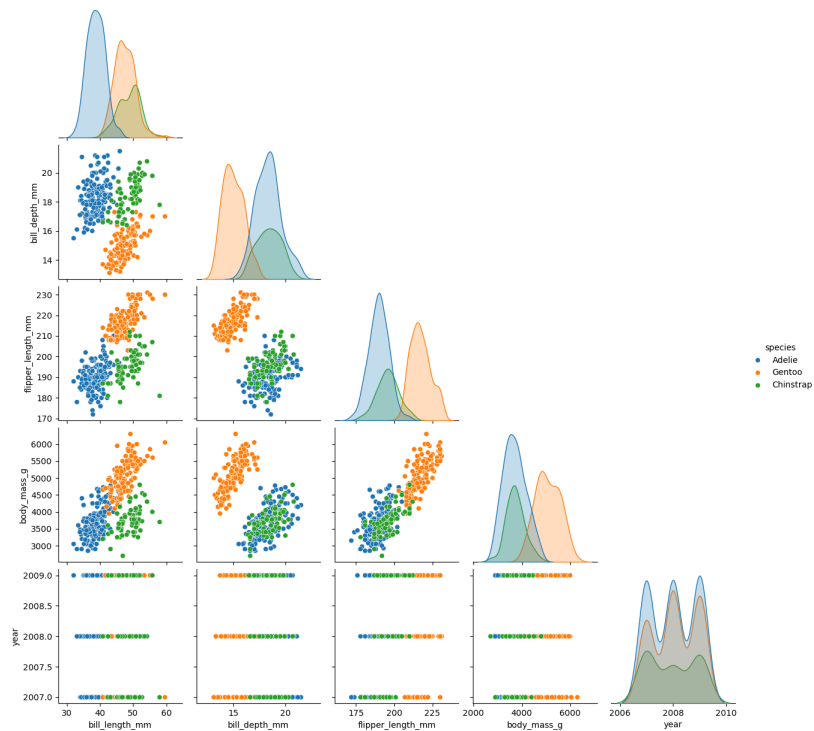


Foundations of Data Science

Lecture 3: Visualization

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Last lecture produced plots like this one. This lecture is about drawing them well.

Two kinds of plots

- **Exploratory**: made for yourself, quickly and in numbers, to see what the data show. That was Lecture 2.
- **Explanatory**: made for others, few and polished, to carry one message.

The building blocks are the same, the standards are not. Later in the course, the same plots come back to show where a model fails (Lecture 7).

Encoding data with visual cues

14, 37, 75

In pairs, try to come up with as many representations/encodings of this "data" as possible.

Visual cues

Visual cues are elements of a visualization that encode data. They are expressed as marks and channels:

- A **mark** is a geometric primitive such as points, lines, or areas.
- A **channel** is an attribute of a mark that can be used to encode data, such as position, size, shape, or color.

➔ Points



➔ Lines



➔ Areas



Marks are geometric primitives.

→ Position

→ Horizontal



→ Vertical



→ Both



→ Color



→ Shape



→ Tilt



→ Size

→ Length



→ Area



→ Volume



Visual channels control the appearance of marks.



(a)



(b)



(c)



(d)

(a) Bar charts encode data using line marks, controlled by a vertical position channel (height) and a horizontal position channel (category). (b) Scatterplots encode data using point marks, controlled by two position channels (x and y). (c) A third variable can be encoded using a color channel. (d) A fourth variable can be encoded using a size channel (area of the point).

Anatomy of a plot

Every plot follows the same recipe, the **grammar of graphics**:

data → marks → channels → scales → coordinates → guides

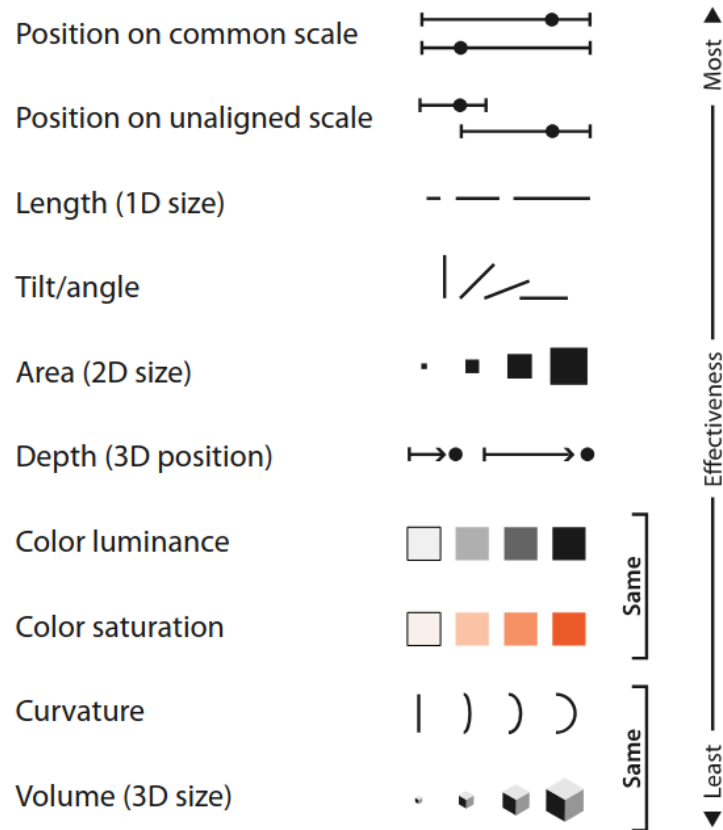
The rest of this lecture takes these pieces in order. Guides are the axes, legends and labels that make the encoding readable.

Two principles

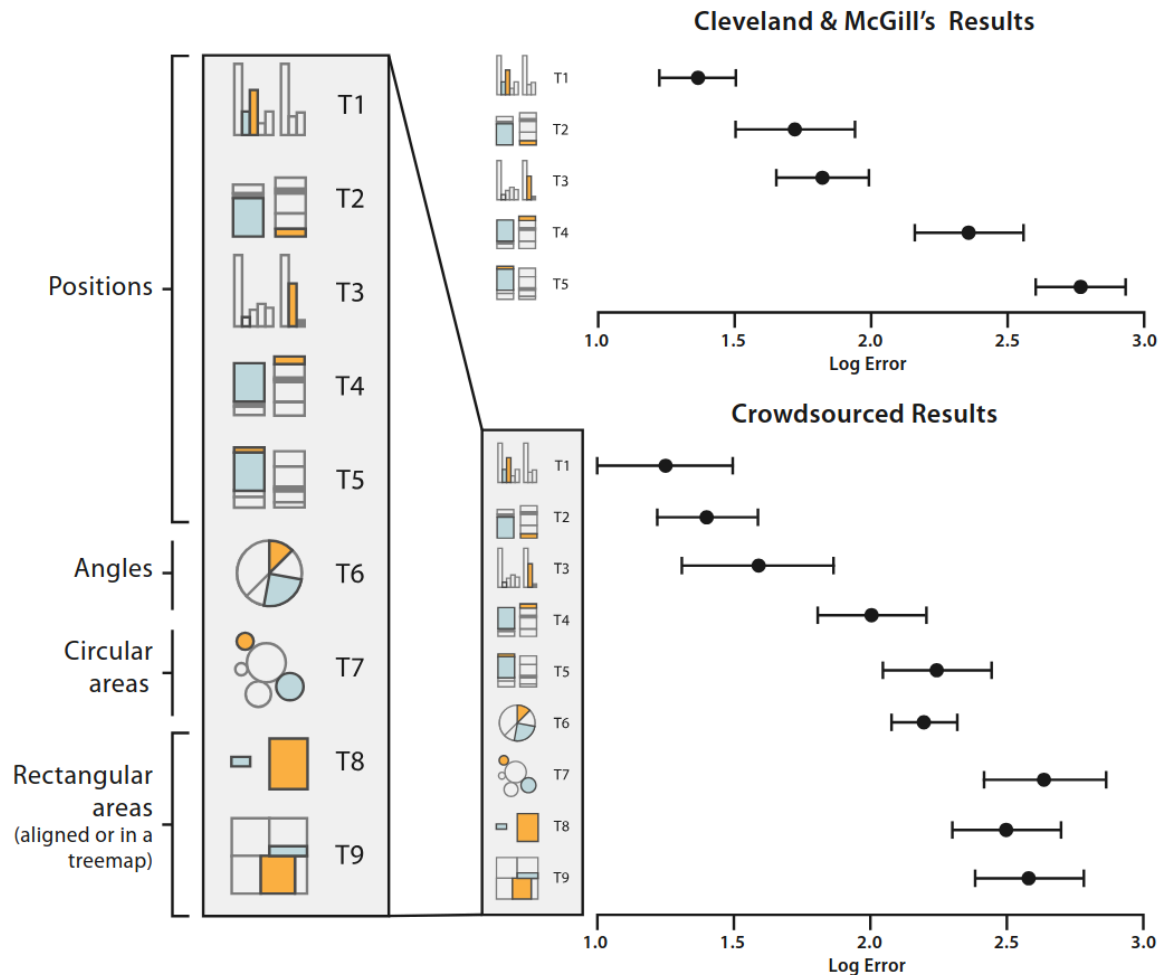
- **Expressiveness:** show all the data facts, and only the data facts.
- **Effectiveness:** encode them with the channels a reader decodes most accurately.

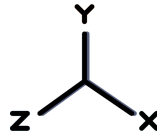
Perceptual hierarchy

Effectiveness is measurable: not all channels are read equally well.



Cleveland and McGill (1984) conducted experiments to evaluate the accuracy of visual channels for encoding quantitative data.

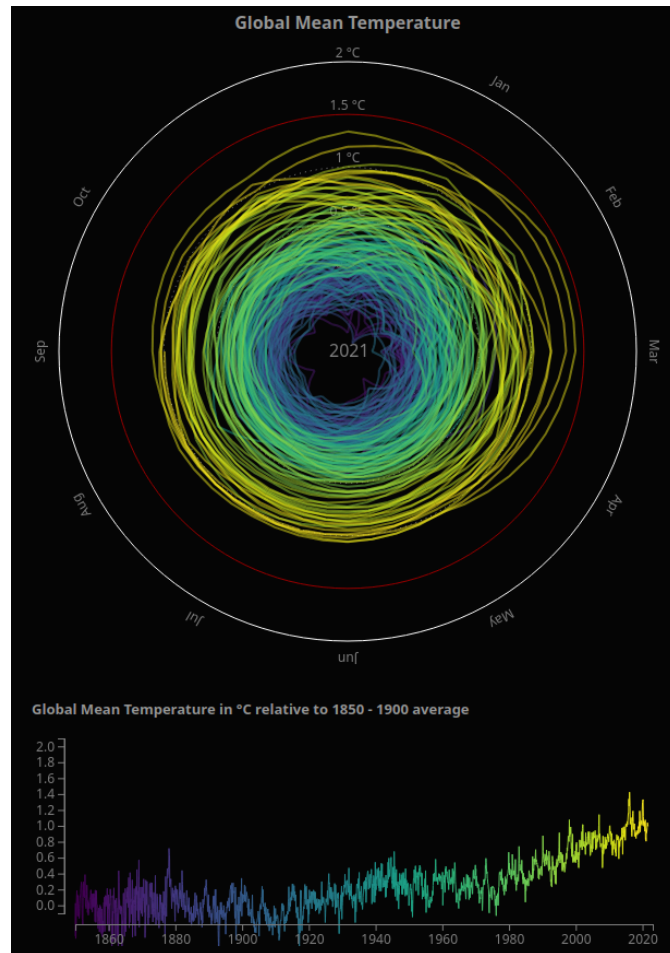




Coordinate systems

Before mapping (numerical) data to visual channels, it is important to choose an appropriate coordinate system as it affects the perception and effectiveness of the visualization.

- **Cartesian coordinates:** intuitive and effective for most data types.
- **Polar coordinates:** useful for cyclic data (e.g., time of day, seasons), but can distort perception of lengths.



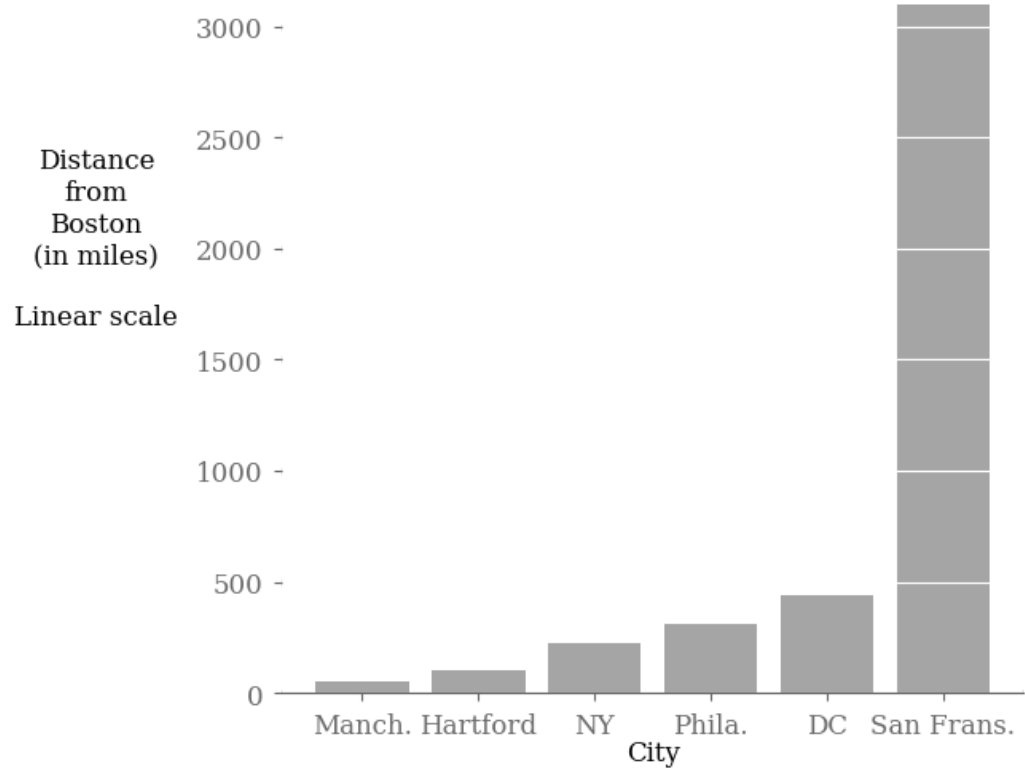
Climate spiral vs. line chart showing global mean temperature change over time.



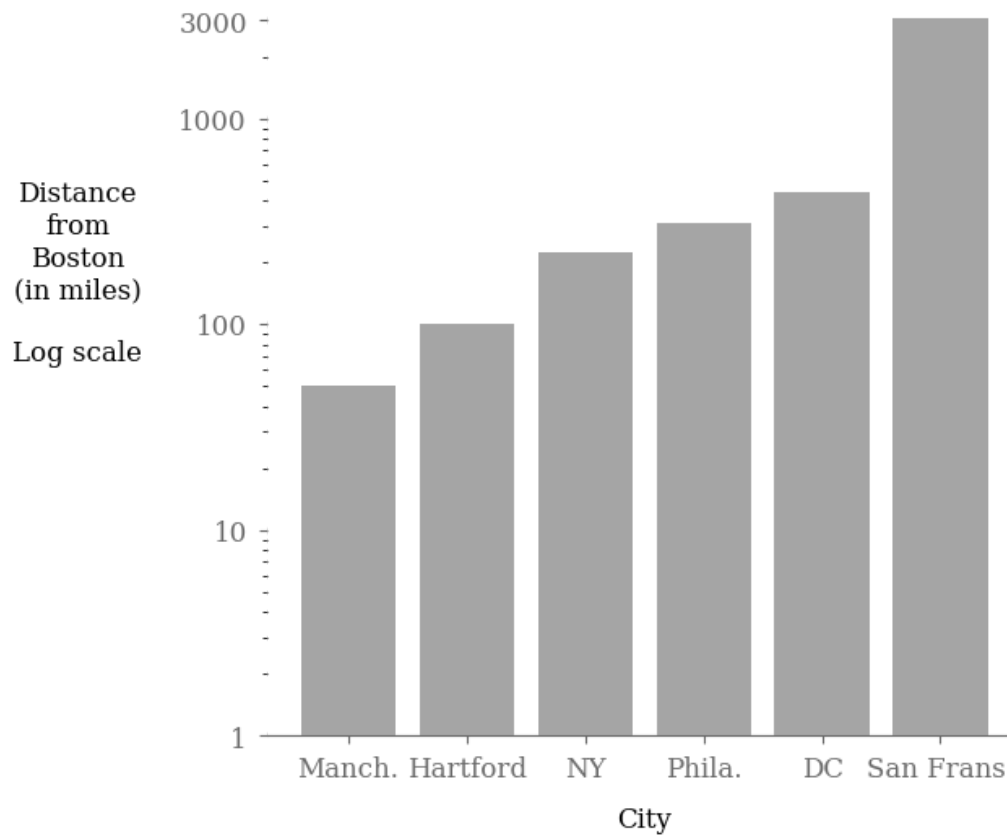
Scales and transformations

Sometimes, data spans several orders of magnitude or has a skewed distribution. In such cases, applying a transformation or using a different scale can improve the interpretability of the visualization.

- **Linear scale:** preserves the original data values.
- **Logarithmic scale:** useful for data spanning several orders of magnitude.
- **Quantile scale:** divides data into bins holding equal numbers of records, useful for skewed distributions.



Linear scales can be dominated by large values, dwarfing smaller values and making it hard to see small variations.



Showing data on a logarithmic scale can prevent large values from dominating the visualization and reveal patterns among smaller values.

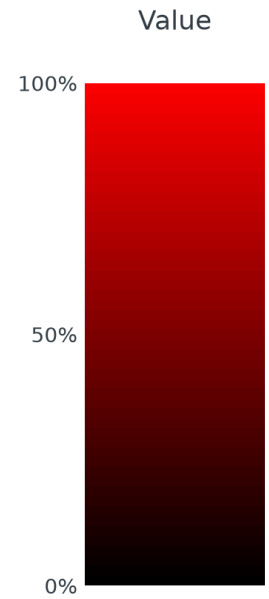
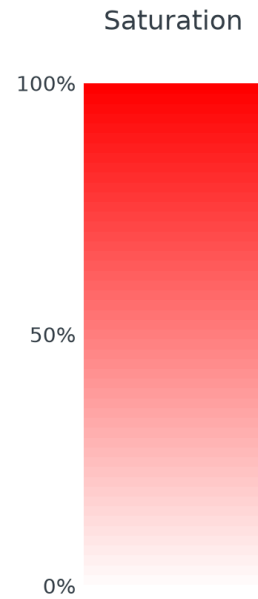


Colors

Color is a powerful channel for encoding categorical and quantitative data.

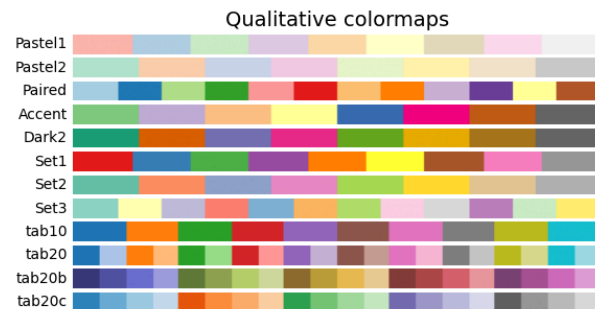
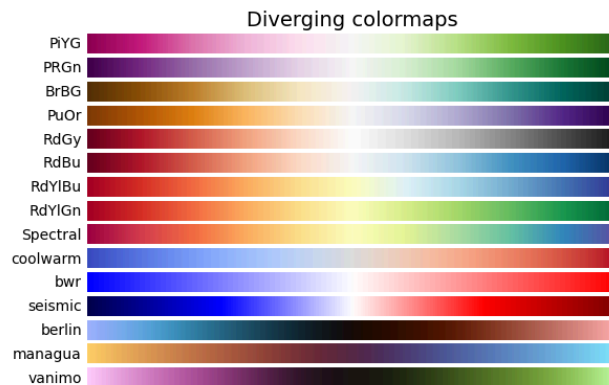
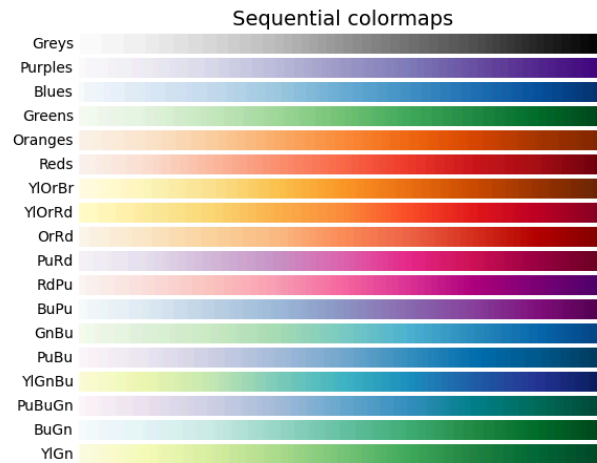
The primary representation system is the Hue, Saturation, Value (HSV) model:

- **Hue**: the type of color (e.g., red, green, blue), numerically represented as an angle on the color wheel (0-360 degrees).
- **Saturation**: the intensity or purity of the color (from gray to full color), represented as a percentage (0-100%).
- **Value**: the brightness of the color (from black to full brightness), represented as a percentage (0-100%).



A colormap specifies a mapping between data values and colors. There are three main types of colormaps:

- **Sequential colormaps:** vary smoothly from light to dark colors, often using a single hue; should be used for representing ordered data.
- **Diverging colormaps:** vary smoothly between two different hues, with a neutral color in the middle; should be used for representing ordered data with a critical midpoint.
- **Categorical colormaps:** consist of distinct colors; should be used for representing categorical data without inherent ordering.



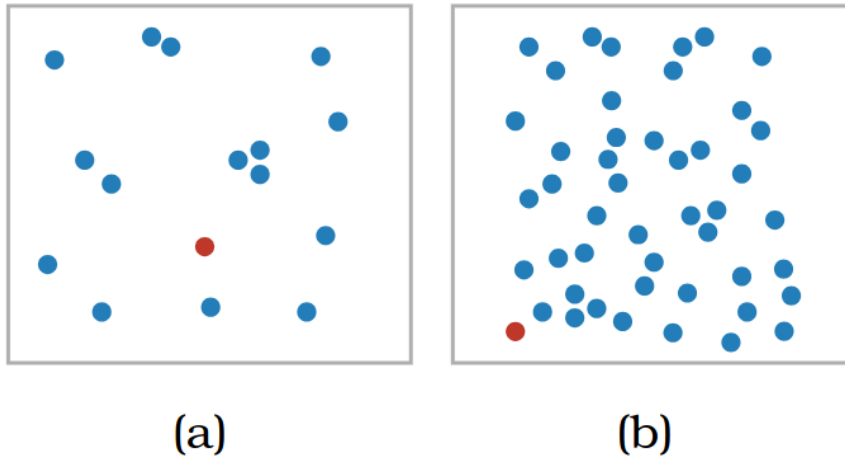


Color vision deficiency affects approximately **1 in 12 men** and **1 in 200 women** worldwide. **Color is not a reliable channel for encoding information.**

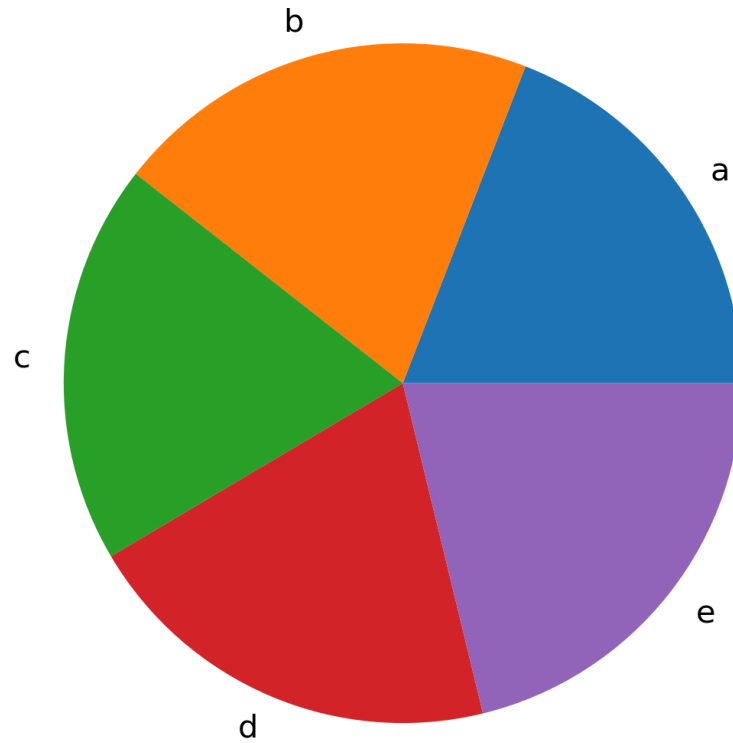


Perceptually uniform colormaps ensure that equal steps in data are perceived as equal steps in color.

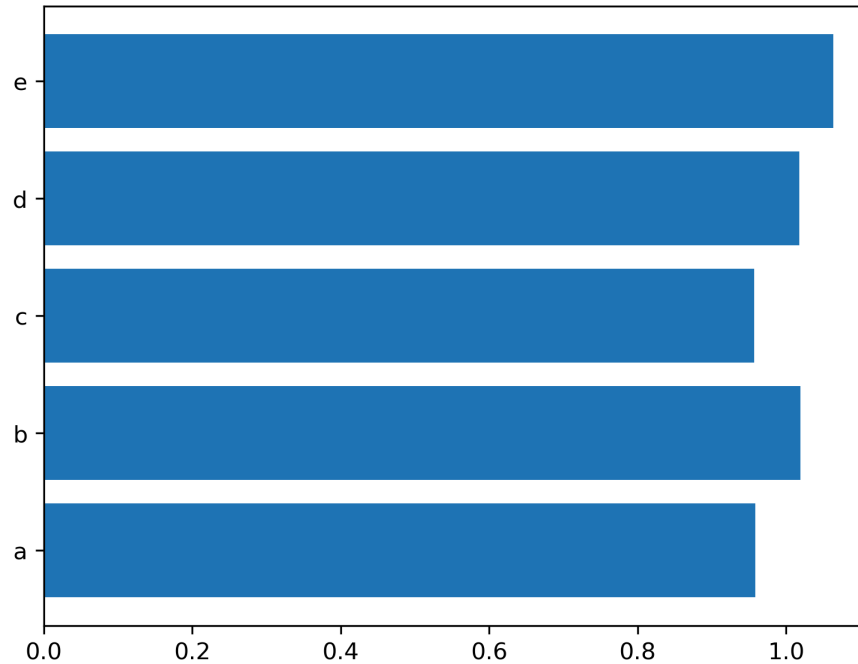
Finally, colors can also be used to **draw attention** to specific elements in a visualization, such as highlighting important data points or trends.



Anti-patterns

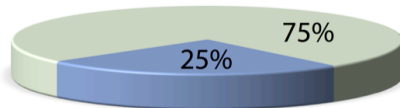


What category is the largest?

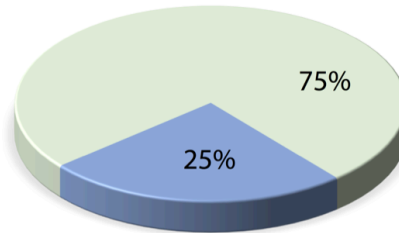


Same data! **Lengths and positions are easier to compare** than angles and areas.

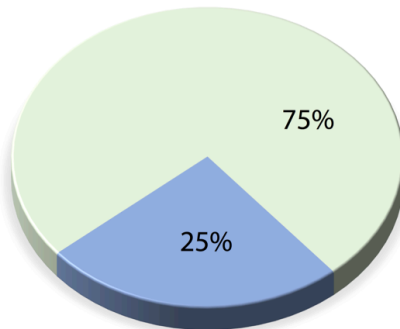
a



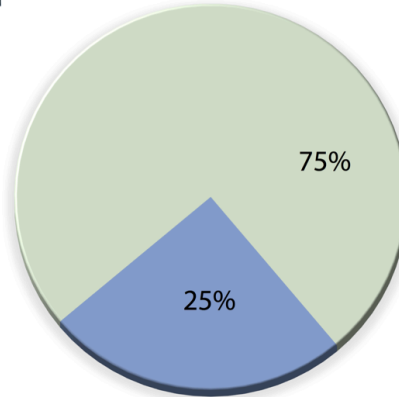
b



c



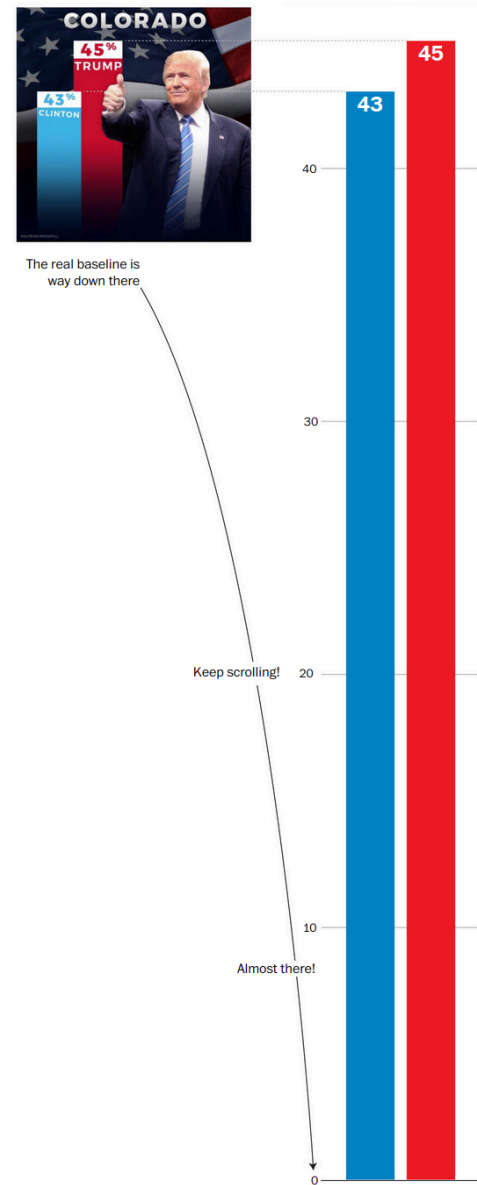
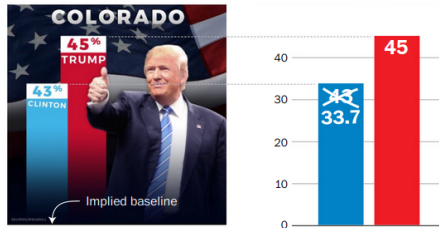
d

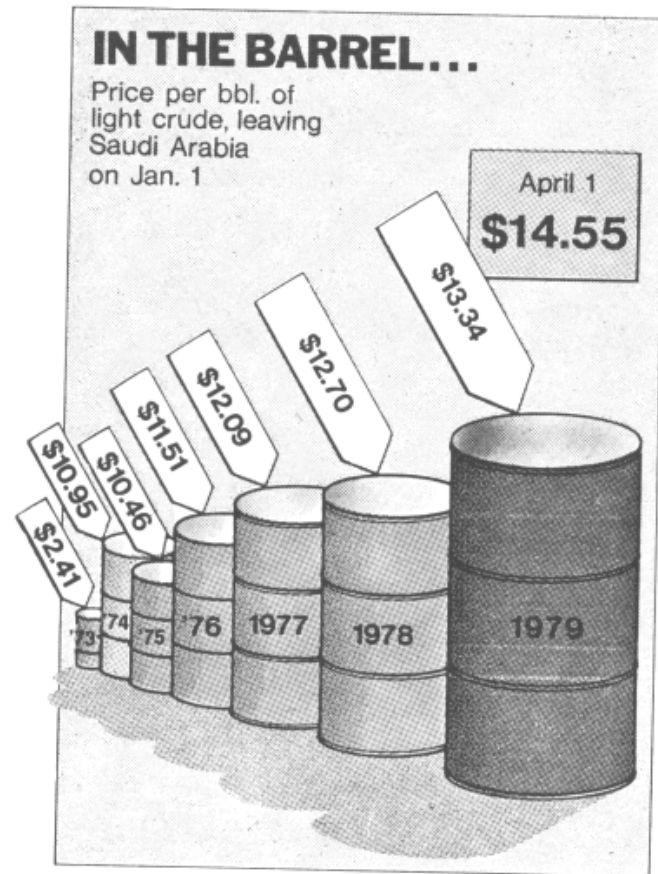
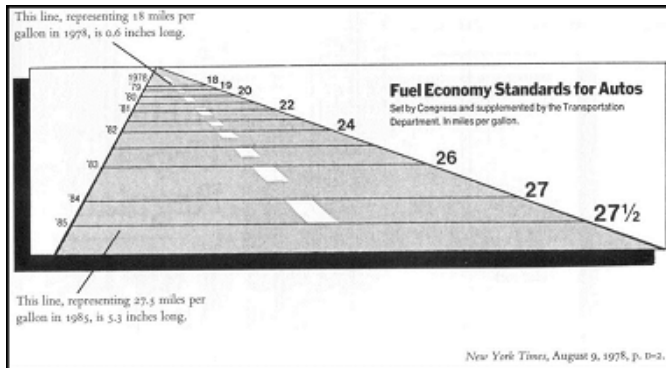


Do not go for 3D. It distorts perception and adds unnecessary complexity.



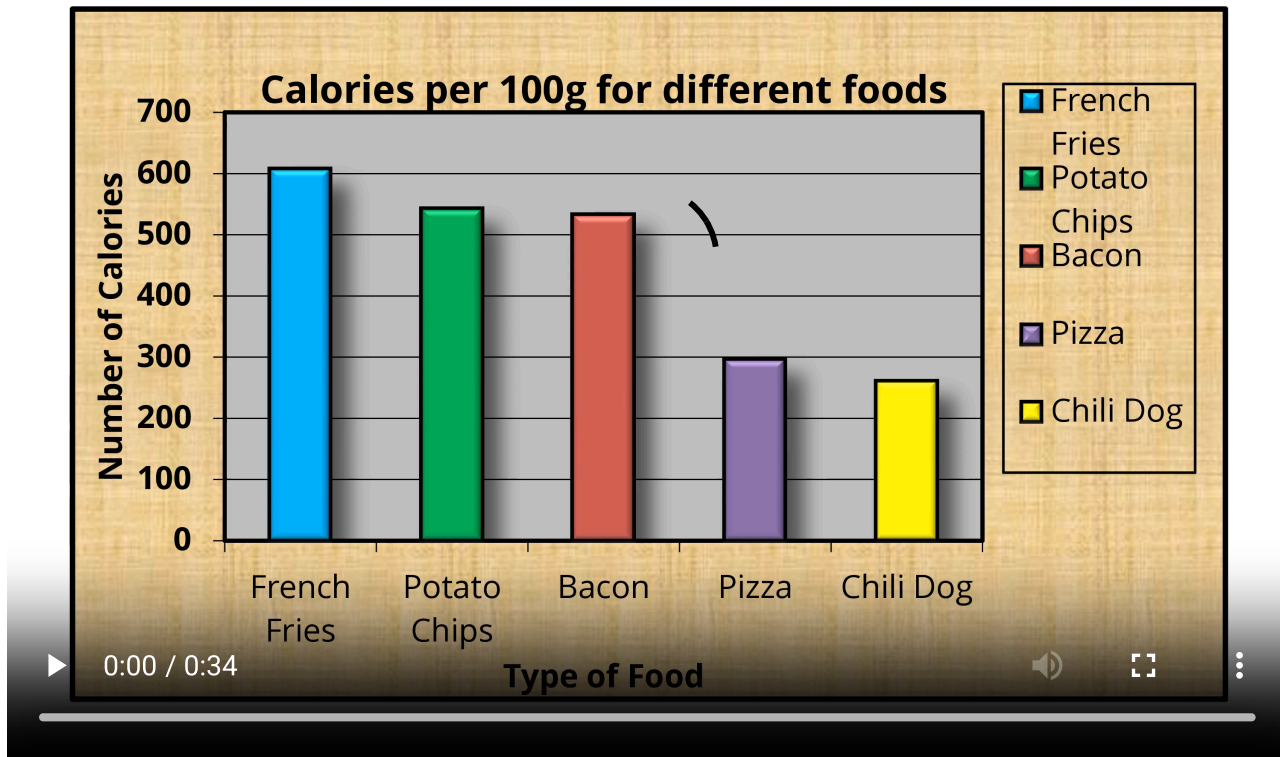
Do not exaggerate reality by truncating axes or using misleading aspect ratios.





Do not lie. Use visual channels that accurately represent the data.

Remove backgrounds

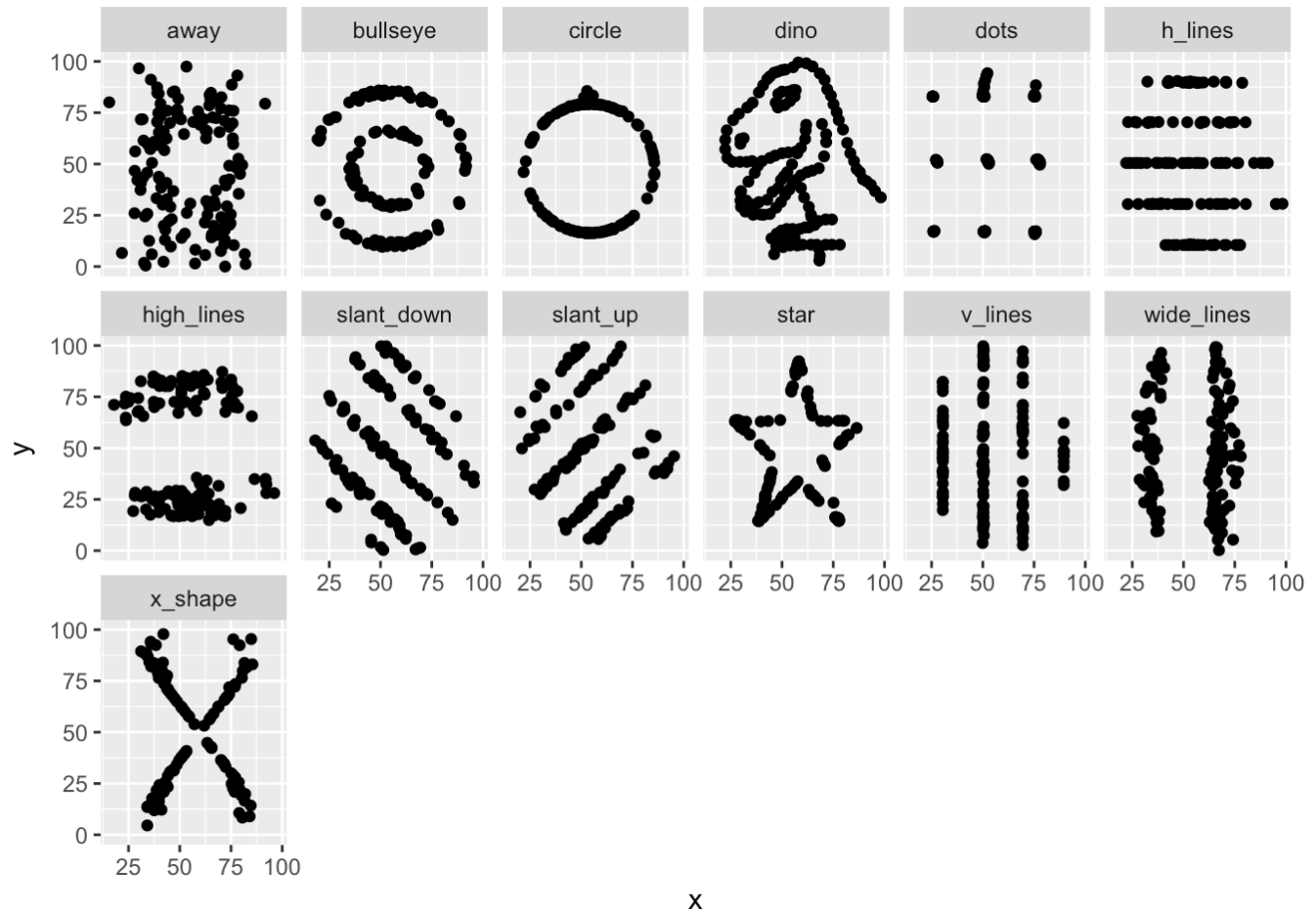


Maximize the data-ink ratio by removing unnecessary elements.

dataset	mean_x	mean_y	sd_x	sd_y	cor
away	54.26610	47.83472	16.76982	26.93974	-0.06412835
bullseye	54.26873	47.83082	16.76924	26.93573	-0.06858639
circle	54.26732	47.83772	16.76001	26.93004	-0.06834336
dino	54.26327	47.83225	16.76514	26.93540	-0.06447185
dots	54.26030	47.83983	16.76774	26.93019	-0.06034144
h_lines	54.26144	47.83025	16.76590	26.93988	-0.06171484
high_lines	54.26881	47.83545	16.76670	26.94000	-0.06850422
slant_down	54.26785	47.83590	16.76676	26.93610	-0.06897974
slant_up	54.26588	47.83150	16.76885	26.93861	-0.06860921
star	54.26734	47.83955	16.76896	26.93027	-0.06296110
v_lines	54.26993	47.83699	16.76996	26.93768	-0.06944557
wide_lines	54.26692	47.83160	16.77000	26.93790	-0.06657523
x_shape	54.26015	47.83972	16.76996	26.93000	-0.06558334

Do not summarize the data without visualizing it.

The Datasaurus dozen: 13 datasets with identical summary statistics.



Summary statistics can be misleading. **Always visualize the rawest data!**

Choosing the right plot

Start from the message

Choose the graph from the question it answers:

- **Comparison** among data,
- **Distribution** of a variable,
- **Correlation** among variables,
- **Evolution** of a variable.

Then choose the design from that question and the number of continuous variables. For subsets, either distinguish them within one panel, or juxtapose panels sharing identical scales.

What that gives

- **Comparison:** horizontal bars, which must start at zero, or dots along a scale, which need not.
- **Distribution:** every point along a scale; a histogram or a box plot when there are too many.
- **Correlation:** a scatter plot, or an array of them beyond two variables.
- **Evolution:** lines against the independent variable; different units go in different panels sharing a scale.

This is one opinionated but consistent take, not the only one.

Choosing the right graph

Principiae



Watch on

Jean-Luc Doumont, "Choosing the right graph", 2017.

Key takeaways

- Visualize data using the most perceptually effective channels (positions, lengths)
- Use appropriate coordinate systems and scales to enhance interpretability.
- Choose colormaps that accurately represent the data and are accessible to all viewers.
- Avoid common pitfalls such as 3D effects, misleading axes, and unnecessary complexity.



Wrap-up exercise

Let us discuss the following examples. For each of them, identify what is good and what could be improved.

(All plots are taken from MSc theses of previous years. Author names have been removed to protect the innocent.)

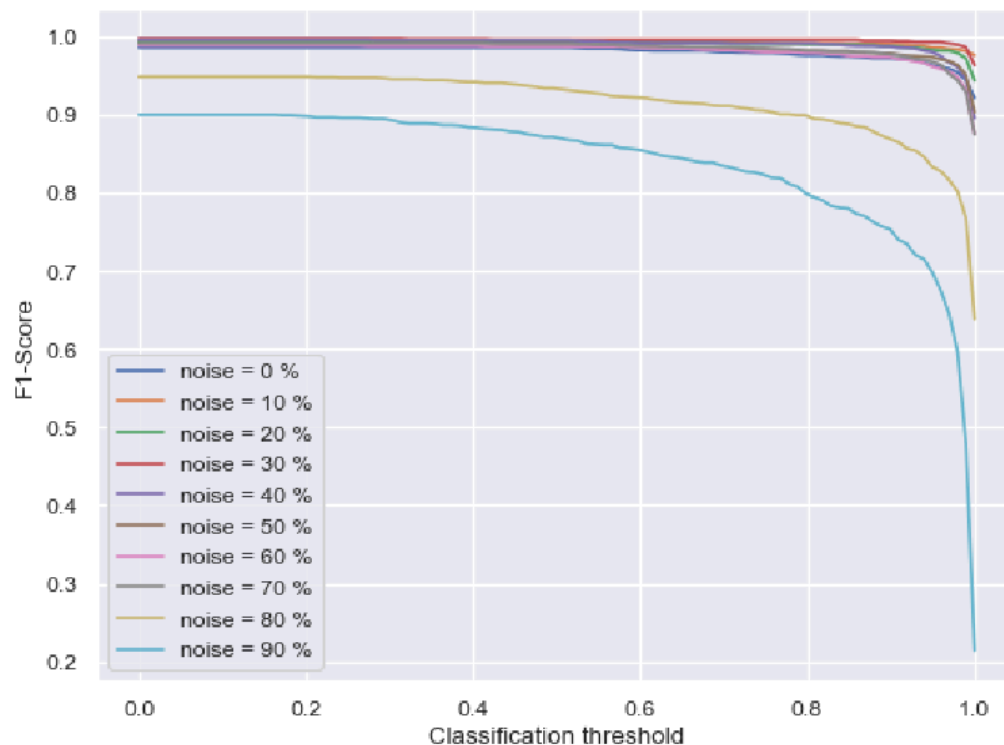


FIGURE 4.21: F1-Score of the best model trained with different quantities of noise going from 0 to 90% and tested on the real conditions dataset according to a classification threshold.

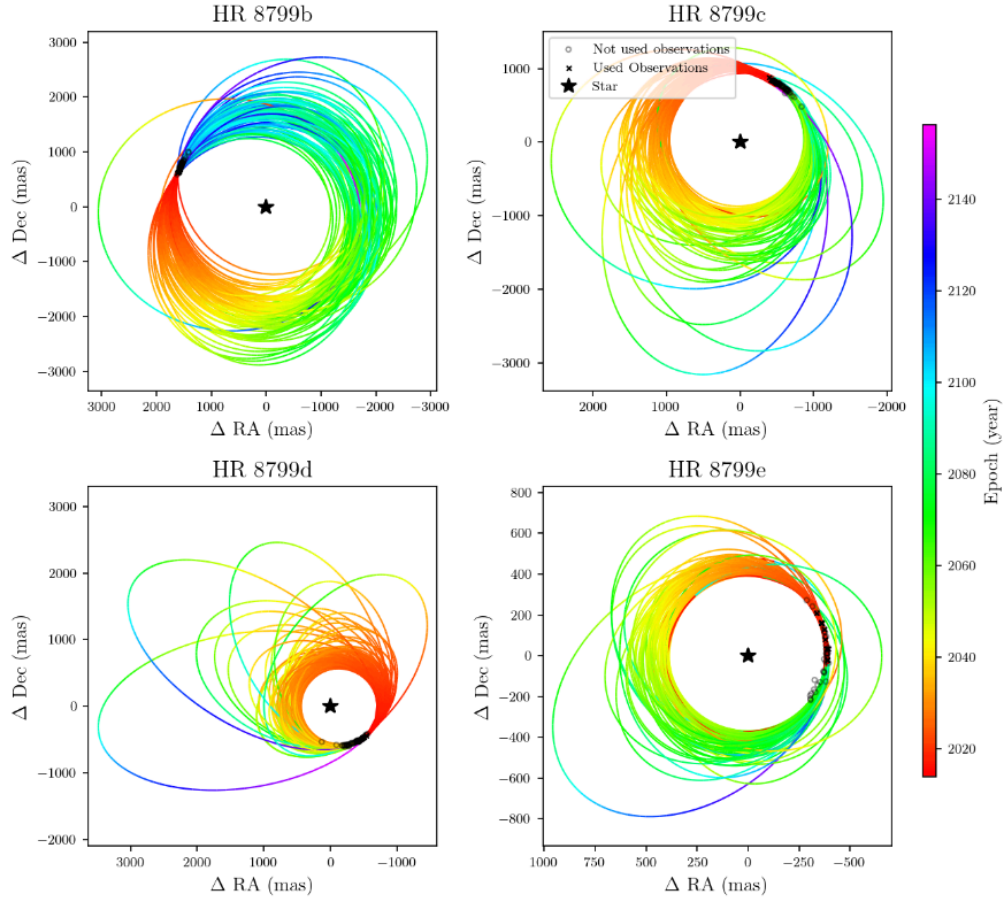
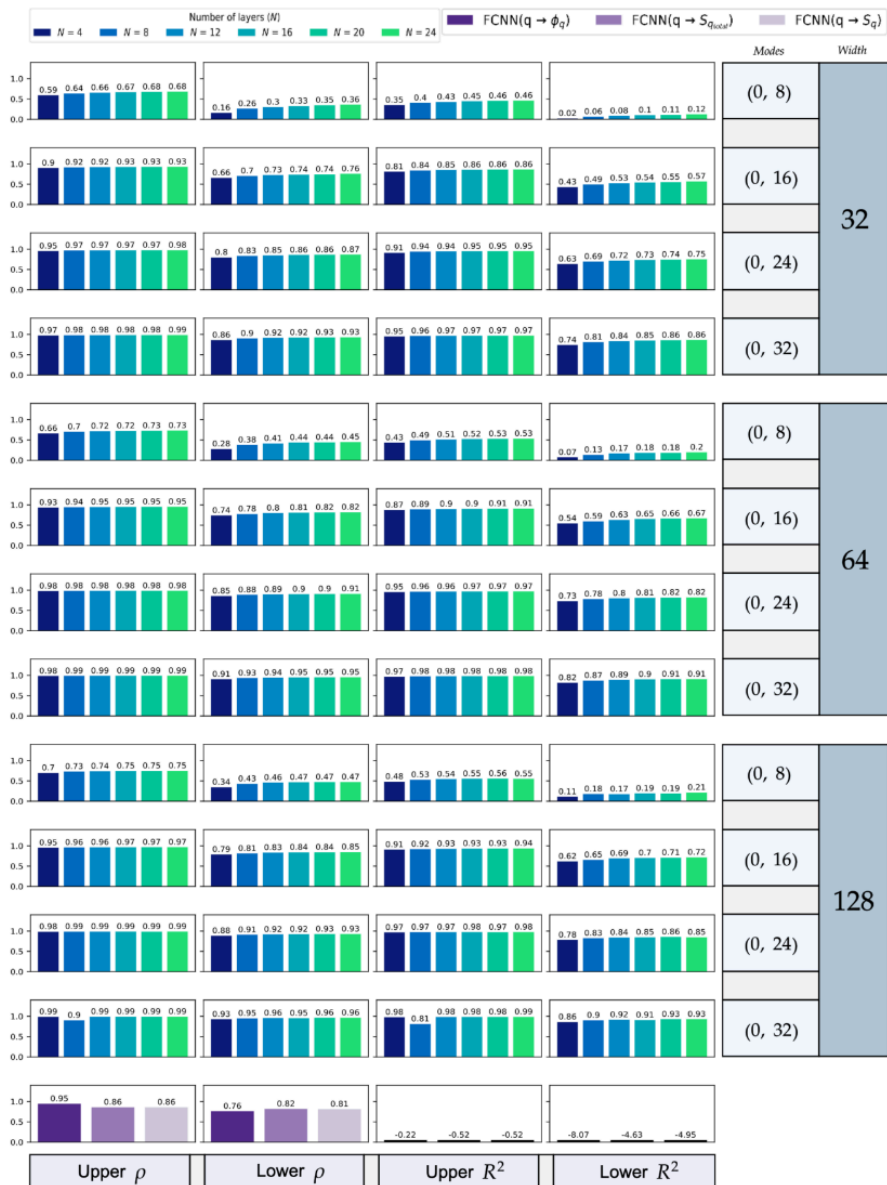


Figure 5.12. Posterior predictive check of the 4 exoplanets of HR 8799, 1000 orbits are generated for each exoplanet. The NPE produces a good approximation of those orbits, however, some impossible orbits are still generated like on HR8799b, HR8799c or HR8799e where there are orbits that are not passing through the observations. The x-axis has been inverted to make the plot comparable with the one in the paper of Sepulveda and Bowler (2022) [40].



The bottom line

A good plot is one that effectively communicates the underlying data and insights to the audience. It should be clear, accurate, and visually appealing, while avoiding unnecessary complexity or distortion of the data.

The end.